

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

As part of the DNS protocol, the **UDP protocol** was used to contact the DNS server to retrieve the IP address for the domain `yummyrecipesforme.com`. The **ICMP protocol** was used to respond with an error message indicating issues contacting the DNS server.

The logs show:

- UDP packets sent to the DNS server on port 53
- ICMP "Destination port unreachable" responses returned
- The error specifically mentions **UDP port 53 unreachable**

This indicates that the DNS server is either down or UDP port 53 traffic is being blocked (most likely by a firewall).

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

The incident occurred around **1:24 p.m.** Multiple customers reported receiving the "destination port unreachable" error when attempting to access the website `www.yummyrecipesforme.com`.

Investigation Steps:

- Loaded tcpdump to capture network traffic while reproducing the issue
- Observed DNS queries using the UDP protocol on port 53
- Received consistent ICMP error responses from the DNS server

Key Findings:

- DNS resolution failed due to UDP port 53 being unreachable
- This prevented the browser from obtaining the web server's IP address, blocking access to the site

Likely Cause: The most probable cause is a **firewall rule blocking UDP port 53 traffic** or a DNS service misconfiguration. It could also result from a targeted attack affecting DNS availability.